

Tutorial 05B Run Report — Concrete Sparsity Exploration

CE 488/5588 — Applied Artificial Intelligence

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This report records the actual run of `concrete-EDA.py` and includes figure outputs generated from the same dataset.

Run Summary

- Script executed: `python3 concrete-EDA.py`
- Captured stdout: `concrete-EDA-run.txt`
- Generated figures: age distribution, strength histogram, correlation plot, and example growth curve

Key Results

- 1,030 cleaned rows, 11 columns
- 14 unique curing ages; 28 days dominates at 41.3%
- 427 unique mixture designs
- 992 unique mixture-age combinations; sparsity ratio 96.3%
- Strongest positive correlation with strength: Cement (+0.498)
- Strongest negative correlation with strength: Water (-0.290)
- No missing values after cleaning; 4 IQR outliers in strength

Figures

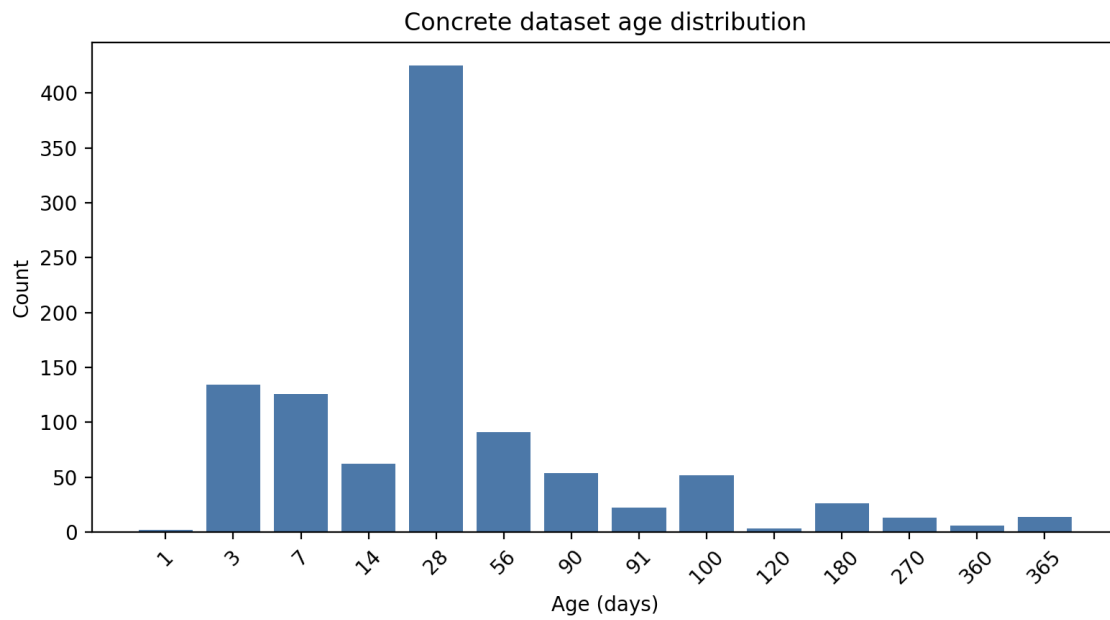


Figure 1: Age distribution in the concrete dataset.

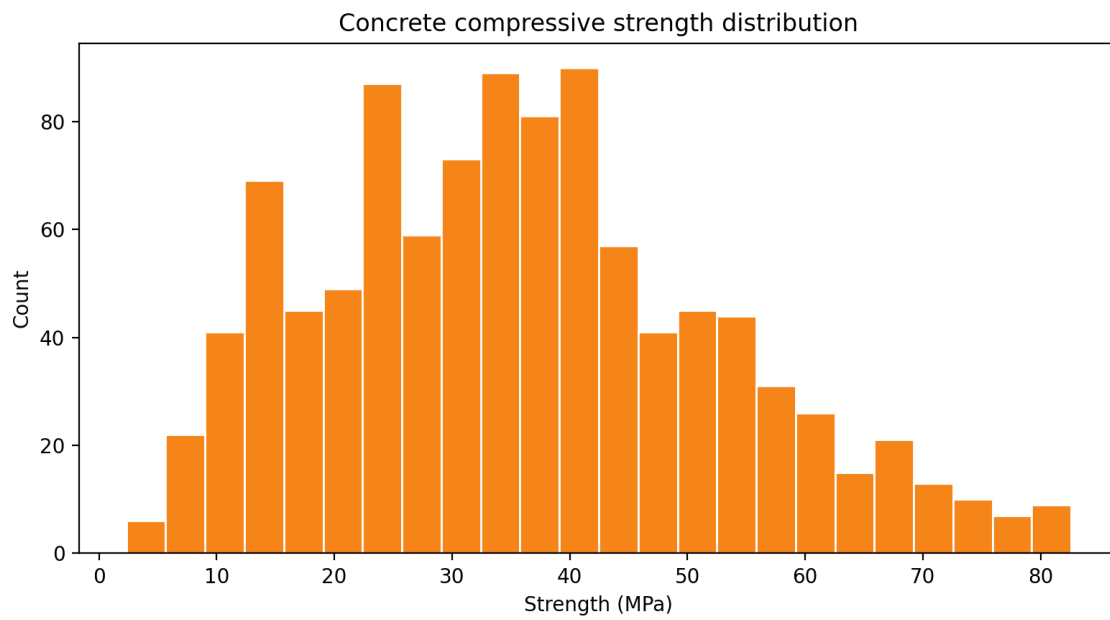


Figure 2: Strength histogram.

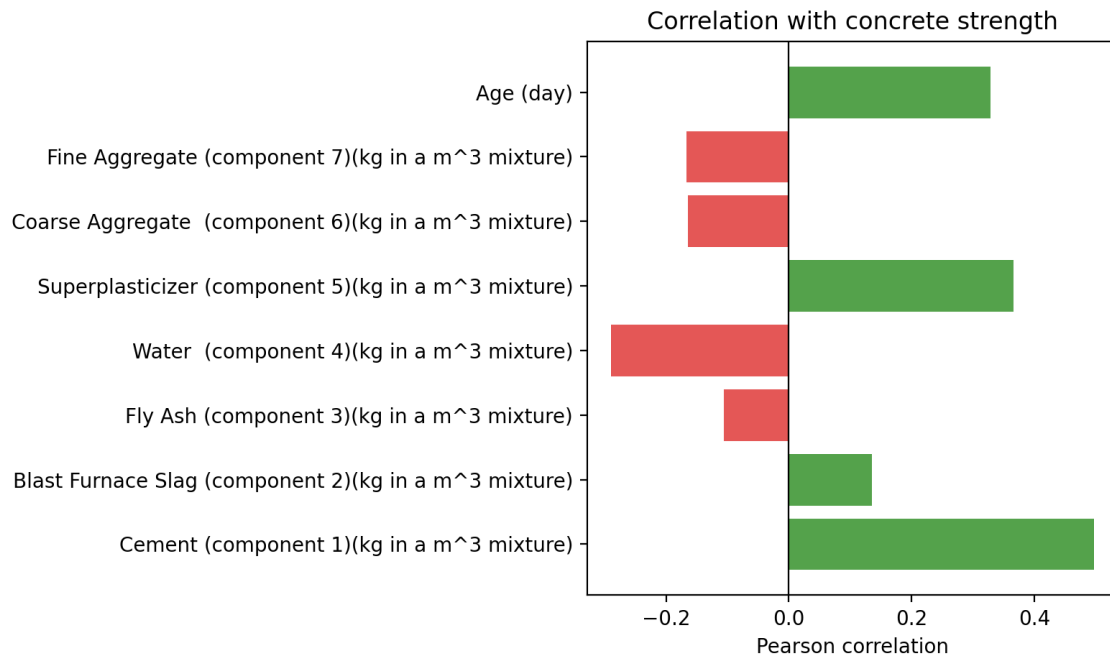


Figure 3: Feature correlation with compressive strength.

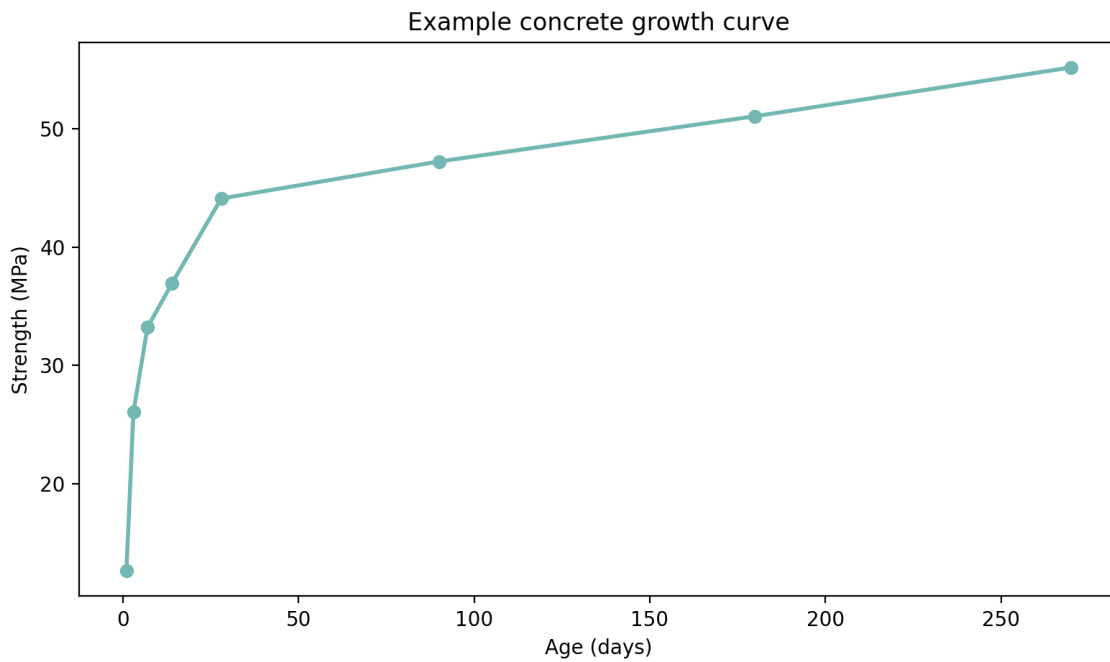


Figure 4: Example growth curve for the mixture with the most age measurements.

Recorded Console Output

```
1  =====
2  CONCRETE COMPRESSIVE STRENGTH - SPARSITY EXPLORATION
3  =====
4
5  Data file: ...
   /Users/chenzhiq/MyProjects/CoursesDev/AppliedAI4Engineering/Tutorials/UCI-Concrete ...
   Data/Concrete_Data_CSV.csv
6  Using encoding: utf-8-sig (handles BOM)
7  After cleaning: 1030 rows
8
9  =====
10  1. BASIC DATASET INFO
11  =====
12  Rows: 1030
13  Columns (11):
14      1. Cement (component 1)(kg in a m3 mixture)
15      2. Blast Furnace Slag (component 2)(kg in a m3 mixture)
16      3. Fly Ash (component 3)(kg in a m3 mixture)
17      4. Water (component 4)(kg in a m3 mixture)
18      5. Superplasticizer (component 5)(kg in a m3 mixture)
19      6. Coarse Aggregate (component 6)(kg in a m3 mixture)
20      7. Fine Aggregate (component 7)(kg in a m3 mixture)
21      8. Age (day)
22      9. Concrete compressive strength(MPa, megapascals)
23      10. f'c at 28 day
24      11. NSC
25
26  Data types:
27      Cement (component 1)(kg in a m3 mixture) float64
28      Blast Furnace Slag (component 2)(kg in a m3 mixture) float64
29      Fly Ash (component 3)(kg in a m3 mixture) float64
30      Water (component 4)(kg in a m3 mixture) float64
31      Superplasticizer (component 5)(kg in a m3 mixture) float64
32      Coarse Aggregate (component 6)(kg in a m3 mixture) float64
33      Fine Aggregate (component 7)(kg in a m3 mixture) float64
34      Age (day) float64
35      Concrete compressive strength(MPa, megapascals) float64
36      f'c at 28 day float64
37      NSC int64
38
39  =====
40  2. BASIC STATISTICS (Numeric Columns)
41  =====
42      Cement (component 1)(kg in a m3 mixture) Blast Furnace Slag (component 2)(kg ...
         in a m3 mixture) Fly Ash (component 3)(kg in a m3 mixture) Water ...
         (component 4)(kg in a m3 mixture) Superplasticizer (component 5)(kg in a ...
         m3 mixture) Coarse Aggregate (component 6)(kg in a m3 mixture) Fine ...
         Aggregate (component 7)(kg in a m3 mixture) Age (day) Concrete ...
         compressive strength(MPa, megapascals)
43 count 1030.00 1030.00 1030.00 1030.00 1030.00 1030.00 1030.00 1030.00 1030.00
44 mean 281.17 73.90 54.19 181.57 6.20 972.92 773.58 45.66 35.82
```

```

45 std 104.51 86.28 64.00 21.35 5.97 77.75 80.18 63.17 16.71
46 min 102.00 0.00 0.00 121.80 0.00 801.00 594.00 1.00 2.33
47 25% 192.38 0.00 0.00 164.90 0.00 932.00 730.95 7.00 23.71
48 50% 272.90 22.00 0.00 185.00 6.40 968.00 779.50 28.00 34.44
49 75% 350.00 142.95 118.30 192.00 10.20 1029.40 824.00 56.00 46.14
50 max 540.00 359.40 200.10 247.00 32.20 1145.00 992.60 365.00 82.60
51
52 =====
53 3. AGE DISTRIBUTION
54 =====
55 Total rows: 1030
56 Unique ages: 14
57 Age (days) Count % Bar
58 -----
59 1.0 2 0.2%
60 3.0 134 13.0% #####
61 7.0 126 12.2% #####
62 14.0 62 6.0% ###
63 28.0 425 41.3% #####
64 56.0 91 8.8% ####
65 90.0 54 5.2% ##
66 91.0 22 2.1% #
67 100.0 52 5.0% ##
68 120.0 3 0.3%
69 180.0 26 2.5% #
70 270.0 13 1.3%
71 360.0 6 0.6%
72 365.0 14 1.4%
73
74 =====
75 4. UNIQUE MIXTURE DESIGNS
76 =====
77 Total rows: 1030
78 Unique mixtures: 427
79 Avg rows per mixture: 2.41
80 Max rows per mixture: 20
81
82 Zero-inflated components (fraction of zeros):
83 Cement (component 1)(kg in a m^3 mixture) 0.0%
84 Blast Furnace Slag (component 2)(kg in a m^3 mixture) 45.7%
85 Fly Ash (component 3)(kg in a m^3 mixture) 55.0%
86 Water (component 4)(kg in a m^3 mixture) 0.0%
87 Superplasticizer (component 5)(kg in a m^3 mixture) 36.8%
88 Coarse Aggregate (component 6)(kg in a m^3 mixture) 0.0%
89 Fine Aggregate (component 7)(kg in a m^3 mixture) 0.0%
90
91 =====
92 5. GROWTH CURVE DATA
93 =====
94 Mixtures with 1 age: 246 (57.6%)
95 Mixtures with 2+ ages: 181 (42.4%)
96 Total growth curve rows: 784
97
98 Top 5 mixtures by number of age measurements:

```

```

99     Ages: [np.float64(1.0), np.float64(3.0), np.float64(7.0), np.float64(14.0), ...
        np.float64(28.0), np.float64(90.0), np.float64(180.0), np.float64(270.0)] ...
        Strengths: [12.64, 26.06, 33.21, 36.94, 44.09, 47.22, 51.04, 55.16]
100    Ages: [np.float64(3.0), np.float64(7.0), np.float64(14.0), np.float64(28.0), ...
        np.float64(90.0), np.float64(180.0), np.float64(365.0)] Strengths: [13.22, ...
        20.97, 27.04, 32.04, 35.17, 36.45, 38.89]
101    Ages: [np.float64(3.0), np.float64(7.0), np.float64(14.0), np.float64(28.0), ...
        np.float64(90.0), np.float64(180.0), np.float64(270.0)] Strengths: [33.8, ...
        42.42, 48.4, 55.94, 58.78, 61.92, 67.11]
102    Ages: [np.float64(3.0), np.float64(7.0), np.float64(14.0), np.float64(28.0), ...
        np.float64(90.0), np.float64(180.0), np.float64(270.0)] Strengths: [41.64, ...
        52.61, 59.76, 67.31, 69.66, 71.62, 74.17]
103    Ages: [np.float64(3.0), np.float64(7.0), np.float64(28.0), np.float64(90.0), ...
        np.float64(180.0), np.float64(360.0)] Strengths: [8.06, 14.59, 28.24, 39.36, ...
        44.21, 44.7]
104
105    =====
106    6. SPARSITY ANALYSIS
107    =====
108    Total rows: 1030
109    Unique mixture-age combos: 992
110    Sparsity ratio: 96.3%
111    Interpretation: 3.7% of data points are unique observations
112
113    Combos with 1 observation: 973
114    Combos with 2 observations: 5
115    Combos with 3+ observations: 14
116    Max observations per combo: 4
117
118    =====
119    7. CORRELATION MATRIX (with Strength)
120    =====
121
122    Feature Correlation
123    -----
124    Cement (component 1)(kg in a m3 mixture) 0.498
125    Blast Furnace Slag (component 2)(kg in a m3 mixture) 0.135
126    Fly Ash (component 3)(kg in a m3 mixture) -0.106
127    Water (component 4)(kg in a m3 mixture) -0.290
128    Superplasticizer (component 5)(kg in a m3 mixture) 0.366
129    Coarse Aggregate (component 6)(kg in a m3 mixture) -0.165
130    Fine Aggregate (component 7)(kg in a m3 mixture) -0.167
131    Age (day) 0.329
132
133    Strongest positive: Cement (0.498)
134    Strongest negative: Water (-0.290)
135
136    =====
137    8. OUTLIER DETECTION (IQR Method)
138    =====
139    Q1: 23.71, Q3: 46.14, IQR: 22.43
140    Lower bound: -9.93, Upper bound: 79.77
141
142    Number of outliers: 4

```

```

143
144     Strength Age Cement
145         82.60 91 389.9
146         81.75 28 315.0
147         80.20 56 323.7
148         79.99 28 540.0
149
150     Strength distribution:
151         Min: 2.33 MPa
152         Max: 82.60 MPa
153         Mean: 35.82 MPa
154         Median: 34.45 MPa
155         70th percentile: 43.29 MPa (70% of data below this)
156
157     =====
158     9. MISSING VALUES
159     =====
160     No missing values detected.
161
162     =====
163     10. DATA IMBALANCE SUMMARY
164     =====
165     Bin Count %
166     -----
167     Early (1-7 days) 262 25.4%
168     Standard (8-28 days) 487 47.3%
169     Medium (29-90 days) 145 14.1%
170     Long (91-180 days) 103 10.0%
171     Very Long (181-365 days) 33 3.2%
172
173     Key insight: Most data is concentrated at 28 days (~41%).
174     This means models may overfit to the 28-day regime and
175     perform poorly on other curing times.
176
177     =====
178     EXPLORATION COMPLETE
179     =====

```